

Date: **Dec 7, 25**

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Assignment # 02

Q:1 Type-1 vs Type-2 Hypervisors (200+VMs)

Ans. Hardware Utilization:

Type-1: Near-native Performance, minimal overhead.

Type-2: Lower efficiency due to host OS.

Latency:

Type-1: Low I/O and scheduling latency.

Type-2: Higher latency

Security:

Type-1: Strong (no host OS attack surface)

Type-2: Weaker (host OS vulnerabilities)

VM Isolation:

Type-1: Strong hardware-level isolation

Type-2: OS-dependent isolation

Recommendation:

Use Type-1 hypervisor: Best for scale, performance, and security in university labs.

Q-2 High availability VM Migration

Ans Host Overload Handling:
DRS detects overload → triggers
VMotion → migrates VMs to less-loaded hosts.

1. When DRS won't migrate:
No available capacity
Affinity rules
DRS disabled/manual
Network
2. Memory pre-copy & Dirty pages
RAM copied while VM runs.
Dirty pages re-copied
High dirty rate → longer migration time.

VMotion Failure case + mitigation

Failure: Network/storage loss during migration

Mitigation: Redundant networks, shared storage,

3. HA restart fallback.

Q.3 VM vs Containers

Ans.

Namespaces & groups:

Containers isolate processes/resources

VMs isolate full OS + kernel

Image layering:

Containers used layered images → faster builds & updates.

Startup:

Containers: ms-seconds

VMs: seconds-minutes

Security boundary

VMs: Strong (separate kernel)

Containers: Weaker (shared kernel)

Load Balancing:

Containers need dynamic LB & service discovery.

VMs scale slower.

When containers perform worse:

High I/O workloads

Strong isolation requirements

Kernel-specific applications

Q.4 MPI All-to-All Communication

Ans 1.

Total Communications

$$\text{Total sends} = P(P-1)$$

2. Scalability issue at $P=512$

~261k messages

Network congestion

High latency → poor scalability

3. Better MPI Approach:

Use MPI_Allgather / MPI_Allreduce

Complexity reduces from $O(P^2)$ → $O(P \log P)$ or $O(P)$

Q.5 MPI Weather Simulation (16 nodes)

Ans 1. Domain Decomposition:

Split grid into 16 sub-domains

Each node processes one block

2. Hello Exchange:

Neighbor boundary data exchanged every timestamp.

Increases communication overhead

3. Network Topology:

Fat-tree: Better for collectives

Torus: Better for neighbor communication

4. Collective Bottleneck:

Higher prediction frequency $\rightarrow$ more global sync $\rightarrow$ performance drop

5. Performance Model:

$$T = T_s + T_c(P)$$

$T_s = W/P$ (parallel compute)

$T_c = \alpha + \beta\sqrt{1/P}$ (latency + hello data)

Communication dominates as P increases